Machine Learning vs. Deep Learning:

Matching the Approach to the Problem in Two Real-World Applications

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Introduction

Machine learning (ML) and deep learning (DL) both enable computers to learn from data, but they differ fundamentally in how features are obtained. In traditional ML, humans manually select the important characteristics of the data and feed them into algorithms such as linear regression, decision trees, or support vector machines. In DL, multi-layer neural networks discover the relevant features automatically from raw input, which makes them powerful for unstructured data but demanding in data volume and computation. This report examines one example of each approach from the Workshop Two lesson: house price prediction as a traditional ML problem and machine translation as a DL problem. For each example, I identify a real-world application, explain why the chosen approach fits the problem, and explain why the opposite approach would be a poor fit.

Example 1: House Price Prediction (Traditional Machine Learning)

The Example and Its Real-World Application

The lesson describes house price prediction as a classic traditional ML task: features such as the number of bedrooms, square footage, and neighborhood quality are fed into a linear regression algorithm to predict a home's price. A large-scale, real-world application of this exact approach is the computer-assisted mass appraisal (CAMA) systems used by county assessor offices to value hundreds of thousands of properties for tax purposes. These systems are built primarily on multiple regression analysis over property characteristics, a practice codified in professional appraisal standards (International Association of Assessing Officers [IAAO], 2017). The same underlying idea also powers consumer-facing automated valuation models such as Zillow's Zestimate, which brought regression-based home valuation to everyday buyers and sellers.

Why Traditional ML Is Suitable

Traditional ML fits this problem for four reasons. First, the data is structured and the informative features are already known. Decades of appraisal practice have established which variables drive price, so manual feature engineering is a strength here rather than a bottleneck: the human knowledge is real and available, and the lesson notes that traditional ML is designed to leverage exactly this kind of human-selected feature set. Second, the data volume is modest. A county records thousands, not billions, of comparable sales, which is more than enough for regression to estimate reliable coefficients (James et al., 2021) but far below the scale at which deep networks thrive. Third, and most decisively, interpretability is a legal requirement, not a preference. Assessed values can be appealed by homeowners and are subject to regulatory review, and a linear model can be defended directly: each coefficient states how much an additional square foot or bedroom contributes to the estimate. Fourth, regression models train in seconds on ordinary hardware, so assessors can re-fit them cheaply for every neighborhood and every tax year.

Why Deep Learning Is Not Suitable

Deep learning is a poor match for this problem precisely where its strengths become irrelevant or harmful. The core advantage of DL, automatic feature discovery from raw data, offers little value when the data is a small tabular set whose meaningful features are already defined; there are no hidden hierarchical patterns, comparable to edges and textures in images, for deep layers to uncover. With only thousands of local sales, a network with millions of parameters is likely to overfit noise rather than generalize, whereas regression's simplicity acts as protection. The opacity of a deep network is an even more serious problem: a valuation that cannot be explained cannot survive a homeowner's tax appeal or an audit, so the black-box nature of DL becomes a legal liability rather than a mere inconvenience. Finally, GPU-based training and inference would add real cost for marginal accuracy gain at best. In short, deep learning would trade away the interpretability this problem legally requires in exchange for a representational power the problem does not need.

A Personal Motivation: China's Housing Downturn and the Limits of Prediction

My choice of this example is also personal. I am from China, where housing prices have been falling in what has become one of the largest property corrections in modern history: new home prices across seventy major cities have declined for thirty-five consecutive months, inflation-adjusted prices have fallen back below their 2005 level, and some cities have reportedly lost more than forty percent of their value from the 2021 peak (Asia Times, 2026; Global Property Guide, 2026). What strikes me most is the asymmetry of who saw it coming. Economists with data and models warned publicly before the downturn began that the sector, which by some measures accounted for nearly twenty-nine percent of China's GDP, had built up price misalignments so severe that an adjustment was inevitable (Rogoff & Yang, 2021). Ordinary families, who held much of their wealth in property, had no realistic way to reach the same conclusion, because the underlying causes, including policy tightening, developer leverage, demographic decline, and regional oversupply, interact in ways that are invisible without data. To me, this is precisely why interpretable prediction models matter in this domain: they do not merely estimate a price, they can show which factors drive the estimate, turning analysis that was once the privilege of professionals into insight ordinary buyers can understand. It is also a sober reminder of every model's limits: a system trained only on historical prices will miss a structural break unless its features capture the policy and credit conditions that cause one.

Example 2: Machine Translation (Deep Learning)

The Example and Its Real-World Application

The lesson presents machine translation as a natural language processing task handled by deep sequence models such as recurrent neural networks and Transformers. The defining real-world application is Google Translate. In 2016, Google replaced its long-standing phrase-based statistical system with an end-to-end neural machine translation system, reporting dramatic reductions in translation errors in human evaluations (Wu et al., 2016). Modern production translation systems, including Google Translate and DeepL, are built on the Transformer architecture, whose self-attention mechanism allows every word in a sentence to weigh its relationship to every other word (Vaswani et al., 2017). These services now translate across more than one hundred languages for a global user base.

Why Deep Learning Is Suitable

Translation is the textbook case for representation learning because its features cannot be written down by hand. The meaning of a word depends on context, word order, idiom, and dependencies that may span an entire paragraph; no human-defined checklist of characteristics can enumerate this. Deep networks solve the problem the lesson highlights as their signature ability: they learn the necessary features directly from raw text. Three further properties seal the match. The input is unstructured and of variable length, which self-attention handles by modeling relationships across the whole sequence at once. The training data exists at enormous scale in the form of web-mined parallel text, which is exactly the regime where deep models keep improving as data grows (Goodfellow et al., 2016). And the output itself is a generated sequence: translation does not assign a label but must produce a new, grammatical sentence of unpredictable length, a task encoder-decoder neural architectures were explicitly designed for.

Why Traditional ML Is Not Suitable

Traditional ML fails at translation for structural reasons, not merely by degree. First, hand-engineered features cannot capture semantics. Features such as word counts, n-grams, or dictionary lookups treat words as fixed symbols, but the same word carries different meanings in different sentences; the word "bank" in a financial report and "bank" beside a river cannot be distinguished by any static feature list. Second, classical algorithms have the wrong output type. Linear regression, logistic regression, and support vector machines map a fixed-length feature vector to a number or a class label; they contain no mechanism for generating an ordered, variable-length sequence of words. Third, shallow models with local features cannot represent long-range dependencies, such as gender agreement or verb placement governed by words far earlier in the sentence. The history of the field is the clearest evidence: pre-neural translation systems built on exactly these engineered-feature techniques plateaued for years with literal, often ungrammatical output, and were retired almost immediately once neural systems demonstrated their step-change in quality (Wu et al., 2016).

Conclusion

Placed side by side, the two examples show that choosing between ML and DL is not a question of which technology is more advanced but of matching the tool to the problem along four dimensions: the structure of the data (tabular versus raw and unstructured), the volume of data available, the interpretability the context demands, and the computational budget. House price prediction sits at one end of every dimension, and machine translation at the other. The Chinese housing correction adds a final lesson: whichever approach is chosen, a model is only as far-sighted as the features it is given. This conclusion matches my professional experience building a large language model system for text-to-SQL conversion, where the decisive factor in performance was never raw model complexity but the quality and selection of the data the model learned from. Sound AI leadership begins with this kind of fit analysis: the most capable technique is not the most appropriate one unless the problem actually calls for what it offers.

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AI Disclosure

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